

cross_classified_models

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Rows: 7,168

Columns: 27

\$ season_year	<dbl> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013~
\$ club_id	<dbl> 11, 11, 11, 11, 11, 11, 11, 11, 11, 11, 11, 11~
\$ player_id	<dbl> 132, 5293, 6710, 7451, 15185, 15378, 15799, 15~
\$ yellow_cards	<dbl> 3, 1, 4, 8, 3, 1, 4, 5, 1, 2, 3, 0, 2, 3, 0, 2~
\$ red_cards	<dbl> 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
\$ goals	<dbl> 3, 0, 3, 6, 16, 0, 12, 0, 0, 0, 0, 0, 7, 16, 0~
\$ assists	<dbl> 1, 0, 0, 6, 12, 1, 13, 0, 0, 1, 0, 0, 5, 13, 0~
\$ minutes_played	<dbl> 904, 90, 3838, 3795, 2610, 132, 4131, 3047, 90~
\$ name	<chr> "Tomas Rosicky", "Sébastien Squillaci", "Per M~
\$ country_of_birth	<chr> "CSSR", "France", "Germany", "Spain", "Poland"~
\$ country_of_citizenship	<chr> "Czech Republic", "France", "Germany", "Spain"~
\$ date_of_birth	<dtm> 1980-10-04, 1980-08-11, 1984-09-29, 1982-03-2~
\$ sub_position	<chr> "Attacking Midfield", "Centre-Back", "Centre-B~
\$ position	<chr> "Midfield", "Defender", "Defender", "Midfield"~
\$ height_in_cm	<dbl> 179, 185, 200, 177, 182, 172, 168, 183, 188, 1~
\$ market_value_ave	<dbl> 2000000.0, 833333.3, 16000000.0, 10000000.0, 2~
\$ Age	<dbl> 33, 33, 29, 31, 28, 32, 29, 28, 29, 27, 30, 28~
\$ Club	<chr> "Arsenal", "Arsenal", "Arsenal", "Arsenal", "A~
\$ Country	<chr> "England", "England", "England", "England", "E~
\$ year_ex_rate	<dbl> 1.33, 1.33, 1.33, 1.33, 1.33, 1.33, 1.33, 1.33~
\$ market_value_ave_usd	<dbl> 2660000, 1108333, 21280000, 13300000, 30590000~
\$ usd_revenue_2024	<dbl> 3581828.4, 1492428.5, 28654627.2, 17909142.0, ~
\$ foreign_players	<dbl> 0.8214286, 0.8214286, 0.8214286, 0.8214286, 0.~
\$ uefa_coefficient	<dbl> 15, 15, 15, 15, 15, 15, 15, 15, 15, 15, 15, 15~
\$ country_of_birth_fixed	<chr> "Czechoslovak Socialist Republic", "France", "~
\$ Region	<chr> "Europe & Central Asia", "Europe & Central Asi~
\$ citizen_of_club_country	<chr> "No", "No", "No", "No", "No", "No", "No", "No"~

Data Structure

Level 1 (time): season_year: season-year within player Level 2 (player): name: players Level 3 (club): club_id: clubs

L1 time-varying: season_year, yellow_cards, red_cards, goals, assists, minutes_played L2 player-invariant: country_of_birth, sub_position, position, height_in_cm L3 club-invariant: season_year

season_year: 2013, 2014, 2015, 2016, 2017, 2018, 2020, 2021, 2022, 2023, 2024.

position: Attack, Defender, Goalkeeper, Midfield.

Region: East Asia & Pacific, Europe & Central Asia, Latin America & Caribbean, Middle East & North Africa, North America, Sub-Saharan Africa.

citizen_of_club_country: YES, NO.

- Check missing values.

season_year	club_id	player_id
0	0	0
yellow_cards	red_cards	goals
0	0	0
assists	minutes_played	name
0	0	0
country_of_birth	country_of_citizenship	date_of_birth
108	103	2
sub_position	position	height_in_cm
0	0	36
market_value_ave	Age	Club
0	2	0
Country	year_ex_rate	market_value_ave_usd
0	0	0
usd_revenue_2024	foreign_players	uefa_coefficient
0	0	0
country_of_birth_fixed	Region	citizen_of_club_country
108	108	103
usd_revenue_2024_log	season_year_c	
0	0	

A tibble: 145 x 29

season_year	club_id	player_id	yellow_cards	red_cards	goals	assists
<dbl>	<fct>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>

1	2013	12	4311	1	0	0	0
2	2013	15	1073	3	0	2	0
3	2013	15	43274	12	0	3	1
4	2013	15	51168	3	0	5	0
5	2013	15	53173	7	0	3	0
6	2013	16	58124	5	0	5	2
7	2013	27	2514	10	0	9	11
8	2013	46	28021	4	0	0	0
9	2013	148	3476	1	0	0	0
10	2013	148	4192	0	0	0	0

```
# i 135 more rows
# i 22 more variables: minutes_played <dbl>, name <chr>,
#   country_of_birth <chr>, country_of_citizenship <chr>, date_of_birth <dtm>,
#   sub_position <chr>, position <fct>, height_in_cm <dbl>,
#   market_value_ave <dbl>, Age <dbl>, Club <chr>, Country <chr>,
#   year_ex_rate <dbl>, market_value_ave_usd <dbl>, usd_revenue_2024 <dbl>,
#   foreign_players <dbl>, uefa_coefficient <dbl>, ...
```

- Intercept only model: We want to model `usd_revenue_2024` because market value is accounting for inflation and converted to 2024 US dollars.

Fit the model

Cross-classified

- The player went to different clubs across years, thus they are cross-clubs, not nested in one club.
- So we use $(1 | \text{club_id}) + (1 | \text{player_id})$ here, not $(1 | \text{club_id}) + (1 | \text{club_id} : \text{player_id})$ here, since the latter one is used only when player are nested in the clubs.

```
# A tibble: 441 x 2
  player_id num_clubs
  <fct>      <int>
1 182712      7
2 45320      5
3 55735      5
4 119296      5
5 282429      5
6 27389       4
7 35564       4
8 37389       4
```

```

9 37666          4
10 39153         4
# i 431 more rows

```

We will now add a fixed effect for year and allow it to vary by club.

- Recall that year has been centered at 0, so that 2013=0, 2024 = 11. This is because `lmer` struggles when predictors are on vastly different scales than the outcome. If `season_year` is recorded as “2020, 2021, 2022,” these numbers are huge compared to the log-revenue variance. This causes the gradient calculation to fail.

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20052.80
BIC	20087.09
Pseudo-R ² (fixed effects)	0.00
Pseudo-R ² (total)	0.79

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.99	0.07	221.31	32.91	0.00
season_year_c	-0.03	0.00	-8.11	6524.03	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.34
club_id	(Intercept)	0.28
Residual		0.71

- We can see that `player_id` accounts for about 76% of market value while `club_id` accounts for 3%.
- we also fit a random slope for year to compare.
 - To estimate a random slope (`1 + year_centered | player_id`), the model needs multiple data points for *every* player to calculate their individual trajectory. But

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.76
club_id	21	0.03

many players only appear in our data for 1 or 2 years, and the model struggles to distinguish the “player slope” from the “residual error,” causing these convergence issues. Clubs, however, have many players and many years of historical market value data so a random slope for clubs is more robust to fit.

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	19993.96
BIC	20041.96
Pseudo-R ² (fixed effects)	0.00
Pseudo-R ² (total)	0.80

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.98	0.09	187.65	24.67	0.00
season_year_c	-0.03	0.01	-3.17	20.11	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.35
club_id	(Intercept)	0.35
club_id	season_year_c	0.04
Residual		0.70

- AIC and BIC values are lower with the random slope of year.

LRT Statistic: 62.841

Parameter Diff: 2

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.75
club_id	21	0.05

Distribution: Mixture of ChiSq(1) and ChiSq(2)

P-Value: < 0.0001

- We use a helper function to print the p-value for the LRT difference, and find a p-value <.0001 and we reject the Null, (with $.5 \times 2(1) + .5 \times 2(2)$ mixture) and conclude that there is strong evidence for club variation in market value trajectories and therefore we should keep this random slope in the model.

Comparison of Model Performance Indices

Name	Model	AIC (weights)	BIC (weights)
fixed_year	lmerModLmerTest	20040.0 (<.001)	20074.3 (<.001)
random_year	lmerModLmerTest	19982.9 (>.999)	20030.9 (>.999)

Adding first level player and club predictors

- We first want to group center minutes played for each for each club by year. We also need to account for the quadratic relationship between age and market value, by centering age by year by club.

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	15641.28
BIC	15730.42
Pseudo-R ² (fixed effects)	0.28
Pseudo-R ² (total)	0.86

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.82	0.07	223.44	25.40	0.00
season_year_c	0.01	0.01	1.75	20.45	0.10
yellow_cards	0.00	0.00	0.61	5306.45	0.54
red_cards	-0.01	0.02	-0.48	4695.40	0.63
goals	0.01	0.00	3.75	5527.58	0.00
assists	0.01	0.00	4.10	5062.36	0.00
minutes_played	0.00	0.00	24.83	5593.88	0.00
age_c	-0.02	0.00	-51.05	6232.32	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.00
club_id	(Intercept)	0.29
club_id	season_year_c	0.03
Residual		0.51

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.74
club_id	21	0.06

We next add first level club predictors

- We want to center player height in inches across years. We also want to center uefa coefficients by year.

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

Adding level 2 predictors

- We want to center height in inches across years and clubs.

AIC	15533.63
BIC	15691.34
Pseudo-R ² (fixed effects)	0.36
Pseudo-R ² (total)	0.86

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.55	0.22	70.89	1590.63	0.00
season_year_c	0.01	0.00	3.60	6703.56	0.00
yellow_cards	-0.00	0.00	-0.07	5229.11	0.94
red_cards	-0.01	0.02	-0.55	4702.43	0.58
goals	0.01	0.00	2.94	5322.45	0.00
assists	0.01	0.00	3.69	4972.49	0.00
minutes_played	0.00	0.00	26.39	5465.39	0.00
age_c	-0.02	0.00	-50.10	6300.98	0.00
height_in_cm_c	0.01	0.00	3.41	1766.08	0.00
foreign_players	0.38	0.10	3.96	5125.48	0.00
uefa_coefficient_c	-0.00	0.00	-0.71	4975.27	0.48
positionDefender	-0.33	0.06	-5.65	1862.91	0.00
positionGoalkeeper	-0.90	0.09	-9.72	1814.23	0.00
positionMidfield	-0.15	0.06	-2.54	1808.07	0.01
RegionEurope & Central Asia	0.29	0.20	1.46	1795.03	0.14
RegionLatin America & Caribbean	0.55	0.21	2.67	1785.15	0.01
RegionMiddle East & North Africa	0.24	0.31	0.79	1766.50	0.43
RegionNorth America	0.21	0.29	0.73	1800.08	0.47
RegionSub-Saharan Africa	0.36	0.22	1.64	1783.30	0.10
citizen_of_club_countryYes	-0.25	0.03	-7.48	6500.43	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.93
club_id	(Intercept)	0.25
Residual		0.52

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.72
club_id	21	0.05

Why This Syntax Works for Cross-Classification

In `lme4`, you do not need special “cross-classified” syntax commands. The structure is defined by how you request the random effects:

1. **Cross-Classification:** By using `(1 | player_id) + (1 | club_id)`, you are telling R that `player_id` is **not** a subset of `club_id`. R treats them as two independent sources of variation that cross over each other (since players move between clubs).
2. **The Random Slope:** By using `(1 + season_year_c | club_id)`, you are specifically testing the hypothesis that **Club A** might be great at increasing player value over time (steep positive slope), while **Club B** might be stagnant (flat slope), even if they start with different intercepts.

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	15509.74
BIC	15681.17
Pseudo-R ² (fixed effects)	0.35
Pseudo-R ² (total)	0.86

1. The Mathematical Equation (LaTeX)

This is a **cross-classified multilevel model** with random intercepts for players and random slopes for clubs.

Let: * t index the **Time** (Season Year). * i index the **Player**. * j index the **Club**.

The log-transformed market value Y_{tij} is modeled as:

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.93	0.23	69.26	941.99	0.00
season_year_c	0.01	0.01	0.77	20.39	0.45
yellow_cards	0.00	0.00	0.70	5187.98	0.49
red_cards	-0.01	0.02	-0.55	4674.31	0.58
goals	0.01	0.00	3.16	5279.82	0.00
assists	0.01	0.00	4.47	4930.09	0.00
minutes_played_c	0.00	0.00	25.97	5391.80	0.00
age_c	-0.02	0.00	-50.27	6274.92	0.00
height_in_cm_c	0.01	0.00	3.50	1769.36	0.00
foreign_players	0.61	0.11	5.73	3350.38	0.00
uefa_coefficient_c	-0.00	0.00	-1.16	4779.52	0.25
positionDefender	-0.32	0.06	-5.52	1852.48	0.00
positionGoalkeeper	-0.89	0.09	-9.53	1805.01	0.00
positionMidfield	-0.14	0.06	-2.43	1802.89	0.02
RegionEurope & Central Asia	0.29	0.20	1.42	1797.26	0.16
RegionLatin America & Caribbean	0.54	0.21	2.60	1788.02	0.01
RegionMiddle East & North Africa	0.23	0.31	0.73	1768.25	0.47
RegionNorth America	0.22	0.29	0.75	1799.53	0.45
RegionSub-Saharan Africa	0.35	0.22	1.55	1786.06	0.12
citizen_of_club_countryYes	-0.24	0.03	-7.28	6542.70	0.00

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.94
club_id	(Intercept)	0.35
club_id	season_year_c	0.03
Residual		0.51

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.70
club_id	21	0.10

$$\begin{aligned}
Y_{tij} = & \beta_0 + \beta_1(\text{Year}_{tij}) \\
& + \underbrace{\beta_2 \text{Goals}_{tij} + \beta_3 \text{Assists}_{tij} + \dots + \beta_k(\text{Age}_{tij} - \bar{\text{Age}}_{tj})}_{\text{Player Fixed Effects (Time-Varying)}} \\
& + \underbrace{\beta_m(\text{Height}_{tij} - \bar{\text{Height}}_{tj}) + \beta_n \text{Foreign}_{tj} + \beta_p(\text{UEFA}_{tj} - \bar{\text{UEFA}}_t)}_{\text{Club/Context Fixed Effects}} \\
& + \underbrace{\beta_q \text{Pos}_i + \beta_r \text{Region}_i + \dots}_{\text{Time-Invariant Controls}} \\
& + \underbrace{u_{0i}}_{\text{Player RE}} + \underbrace{v_{0j} + v_{1j}(\text{Year}_{tij})}_{\text{Club RE}} + \epsilon_{tij}
\end{aligned}$$

Where the Random Effects are distributed as:

$$\begin{aligned}
u_{0i} & \sim N(0, \sigma_{\text{player}}^2) && \text{(Player Intercept)} \\
\begin{pmatrix} v_{0j} \\ v_{1j} \end{pmatrix} & \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_{\text{club_int}}^2 & \sigma_{\text{cov}} \\ \sigma_{\text{cov}} & \sigma_{\text{club_slope}}^2 \end{pmatrix} \right) && \text{(Club Intercept \& Slope)} \\
\epsilon_{tij} & \sim N(0, \sigma_{\epsilon}^2) && \text{(Residual)}
\end{aligned}$$

2. Interpretation of the Final Model

Here is how you interpret the specific coefficients, especially given the custom centering you applied.

A. Fixed Effects (The “Average” Effects)

- **Intercept (β_0):** The expected log market value for a baseline player (likely a Forward/Midfielder from the reference Region) who has **average performance**, is of **average age relative to their team**, and plays for an **average club** at the starting year ($t = 0$).
- **Season Year (β_1):** This is the **global trend**. It tells you the average annual growth rate in player market value across the entire dataset, holding performance constant. A positive value means the market is inflating generally.
- **Player Performance (Goals, Assists, etc.):** These are standard interpretations. For every 1 additional goal scored, market value increases by β_2 (or $\approx \beta_2 \times 100\%$ since the outcome is logged), holding all else constant.

- **Age_{tij} – Age_{tj} (Group-Mean Centered): Crucial Distinction:** Because you centered by club and year, this does *not* measure the effect of getting older in general. It measures the effect of being **older relative to your teammates**.
 - *Positive Coeff:* “Being the veteran on the team increases your value.”
 - *Negative Coeff:* “Being older than your average teammate decreases your value” (likely true for soccer players past their prime).
- **UEFA_{tj} – UEFA_t (Year-Centered):** This measures the prestige of the club **relative to other clubs that season**. By centering by year, you removed the “grade inflation” of UEFA coefficients over time. A positive coefficient means playing for a club that is currently outperforming the European average boosts a player’s value.

B. Random Effects (The “Variance” Components)

- **Player Random Intercept (u_{0i}):** This captures **innate player quality** (“Superstar Effect”). Even if two players have the exact same goals, assists, and age, Player A might consistently have a higher market value than Player B due to unmeasured factors (marketing potential, leadership, brand).
 - *High σ^2_{player} :* Suggests market value is driven heavily by who the player *is*, not just what they did this year.
- **Club Random Intercept (v_{0j}):** This captures the **baseline club premium**. Playing for Real Madrid or Man City likely adds a baseline value to a player regardless of their performance.
- **Club Random Slope (v_{1j}):** This is the most interesting part. It captures **Club Development Trajectories**.
 - A club with a *high positive random slope* ($v_{1j} > 0$) is an “incubator” or “value creator.” Players at this club see their market value rise faster over time than the global average.
 - A club with a *negative random slope* ($v_{1j} < 0$) might be a place where player values tend to stagnate or decline faster than the market.
- **Covariance (σ_{cov}):** This tells you the relationship between the Club Intercept and Slope.
 - *Positive Correlation:* Rich clubs (high intercept) are also better at growing player value (high slope).
 - *Negative Correlation:* Rich clubs start high but grow slowly (diminishing returns), while smaller clubs start low but grow player value quickly (development clubs).

If we want to back log transform we may need to use the broom package.

```
# A tibble: 6 x 2
  usd_revenue_2024 fitted_original_scale
      <dbl>          <dbl>
1      3581828.      3829181.
2      1492428.      1731085.
3     28654627.     27224914.
4     17909142.     18506097.
5     41191027.     52717810.
6      6715928.      6141327.
```

	M1	M2	M3
(Intercept)	15.981 ***	15.816 ***	15.547 ***
	[15.814, 16.148]	[15.678, 15.955]	[15.118, 15.977]
season__year__c	-0.029 **	0.011	0.010 ***
	[-0.047, -0.011]	[-0.001, 0.023]	[0.005, 0.016]
yellow__cards		0.002	-0.000
		[-0.005, 0.009]	[-0.007, 0.007]
red__cards		-0.012	-0.014
		[-0.060, 0.036]	[-0.062, 0.035]
goals		0.008 ***	0.007 **
		[0.004, 0.012]	[0.002, 0.011]
assists		0.012 ***	0.011 ***
		[0.006, 0.018]	[0.005, 0.017]
minutes__played		0.000 ***	0.000 ***
		[0.000, 0.000]	[0.000, 0.000]
age__c		-0.021 ***	-0.020 ***
		[-0.022, -0.020]	[-0.021, -0.020]
height__in__cm__c			0.012 ***
			[0.005, 0.019]
foreign__players			0.383 ***
			[0.194, 0.572]
uefa__coefficient__c			-0.001
			[-0.003, 0.002]
positionDefender			-0.327 ***
			[-0.440, -0.213]
positionGoalkeeper			-0.902 ***
			[-1.084, -0.720]
positionMidfield			-0.145 *
			[-0.257, -0.033]
RegionEurope & Central Asia			0.294
			[-0.101, 0.689]
RegionLatin America & Caribbean			0.550 **

		M4
(Intercept)		15.933 ***
		[15.482, 16.384]
season__year__c		0.005
		[-0.008, 0.018]
yellow__cards		0.002
		[-0.004, 0.009]
red__cards		-0.014
		[-0.062, 0.035]
goals		0.007 **
		[0.003, 0.011]
assists		0.014 ***
		[0.008, 0.019]
minutes__played__c		0.000 ***
		[0.000, 0.000]
age__c		-0.020 ***
		[-0.021, -0.020]
height__in__cm__c		0.013 ***
		[0.006, 0.020]
foreign__players		0.607 ***
		[0.399, 0.814]
uefa__coefficient__c		-0.001
		[-0.004, 0.001]
positionDefender		-0.322 ***
		[-0.437, -0.208]
positionGoalkeeper		-0.894 ***
		[-1.078, -0.710]
positionMidfield	15	-0.140 *
		[-0.254, -0.027]
RegionEurope & Central Asia		0.289
		[-0.111, 0.689]
RegionLatin America & Caribbean		0.542 **